



Roll No. CS-62

NATIONAL UNIVERSITY OF MODERN LANGUAGES ISLAMABAD
FACULTY OF ENGINEERING AND COMPUTING
DEPARTMENT OF COMPUTER SCIENCE
FINAL-TERM EXAMINATION – SPRING 2025 – BSCS VII A&B (MORNING)

Subject: Computer Graphics

Allowed Time: 03 Hours

Instructor: Sabina Irum

Total Marks: 50

Instructions: -

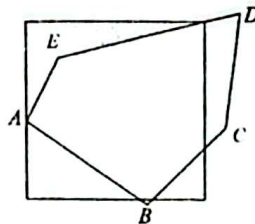
- i. Attempt all questions and return the question paper with the answer sheet.
- ii. Clearly mention the question number before attempting it

QUESTION 1 [CLO-2, C3, PLO-3]

Part 1:

[30 Marks]

1. Given the polygon in the figure, identify how the Sutherland-Hodgman algorithm determines which vertices to keep and which to discard during the clipping process. Provide a step-by-step walkthrough of how the algorithm clips the polygon against a rectangular window, and final result. (NOTE that from 1).



2. In a digital whiteboard application, a user draws a line from point A(9, 18) to point B(14, 22). The system must choose the most efficient algorithm to render this line on a pixel grid. As a computer graphics student, identify which algorithm is more suitable: DDA or Bresenham? Use the chosen algorithm to compute the intermediate points between A and B.
3. A CAD-based drawing tool receives input to draw a line segment from point P(4,1) to point Q(6,4). The application includes a rectangular clipping window defined by vertices A(3,2), B(7,2), C(7,6), and D(3,6). Apply the Cohen-Sutherland line clipping algorithm to this line segment. Plot all relevant points and document each step taken to determine the final visible portion of the line within the clipping window.

QUESTION 2 [CLO-3, C3, PLO-4]

Part 2:

1. **Organize** a C++ program for the following problem in which you are developing a paint application that includes a polygon filling tool. The tool should fill the interior of a convex or concave polygon by starting from an interior pixel and checking all connected neighbors (up, down, left, right, and all diagonals) to ensure the fill stays within the polygon's boundary. Describe and justify which algorithm you would use to fill the polygon. **(10 Marks)**

2. Given a point P(4.4, 2.2) in world coordinates and the following:
Window: $X_{wmin} = 20$, $X_{wmax} = 80$, $Y_{wmin} = 40$, $Y_{wmax} = 80$
Viewport: $X_{vmin} = 30$, $X_{vmax} = 60$, $Y_{vmin} = 40$, $Y_{vmax} = 60$

Construct the window-to-viewport mapping formula to calculate the mapped coordinates (S_x , S_y) of point P. **(5 Marks)**

3. **Construct** a C++ program in OpenGL code in C that sets up a basic scene with both a light source and a material with the following properties i.e. Ambient: RGB = (0.2, 0.2, 0.2) Diffuse: RGB = (0.8, 0.5, 0.3) Specular: RGB = (1.0, 1.0, 1.0) Shininess: 50.0.

Ensure that:

- a. Lighting is enabled, and a light source is configured at position (0.0, 0.0, 10.0).
- b. Depth testing is enabled to ensure correct object rendering.
- c. The background color is set to dark gray (RGB = 0.1, 0.1, 0.1).
- d. The material properties (ambient, diffuse, specular, and shininess) are set for objects in the scene.

(5 Marks)